

Differential Convergence Rates Cause Minority Group Curvature Disparity

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Abstract

Neural networks trained with empirical risk minimization on group-imbalanced data exhibit poor worst-group accuracy. Prior work established that minority groups occupy sharper regions of the loss landscape (sharpness ratio $SR > 1.0$), but the origin of this curvature asymmetry remained unclear. This work investigates the Differential Convergence Rate hypothesis: majority groups converge faster during training, smoothing curvature in majority-relevant parameter directions while leaving minority directions sharp.

Two findings are presented. First, sharpness ratio equals approximately 1.0 at random initialization ($SR_0 = 0.9999$, 95% CI [0.9953, 1.0046], $n = 5$ seeds), indicating that intrinsic data geometry does not cause curvature asymmetry. Second, gradient ratio decay temporally precedes sharpness ratio divergence by $\tau = 3.67$ epochs (95% CI [3.0, 4.0], $n = 3$ seeds). This positive lag is consistent with a causal mechanism wherein majority convergence precedes minority curvature increase.

These findings characterize a diagnostic window: gradient dynamics signal impending curvature divergence 3-4 epochs before it manifests. A proposed update-norm parity intervention was implemented and code-validated but not experimentally executed due to hardware constraints.

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

Deep neural networks trained with empirical risk minimization (ERM) on datasets with spurious correlations systematically underperform on minority groups. On Waterbirds, where 95% of waterbirds appear on water backgrounds, ERM achieves approximately 97% average accuracy but only approximately 75% on the minority group (waterbirds on land). This accuracy gap has motivated interventions including Group DRO, last-layer retraining, and various reweighting schemes.

These methods address the symptom of poor minority accuracy without explaining the mechanism. Recent work established that minority groups occupy sharper regions of the loss landscape, with sharpness ratios $SR > 1.0$. Whether this curvature asymmetry is intrinsic to the data or emerges from training dynamics remained an open question.

This work investigates the Differential Convergence Rate (DCR) hypothesis: majority groups, having more samples, converge faster during training. Their gradient norms decay earlier, smoothing curvature in majority-relevant parameter directions. Minority-relevant directions, receiving fewer updates, remain sharp.

Contributions:

1. Establishment that $SR \approx 1.0$ at random initialization ($SR_0 = 0.9999$, $SEM = 0.0017$), indicating curvature asymmetry is training-induced rather than data-intrinsic.
1. Demonstration that gradient ratio decay temporally precedes sharpness ratio divergence by $\tau = 3.67$ epochs (95% CI [3.0, 4.0]), consistent with a causal relationship.
1. Identification of a 3-4 epoch diagnostic window between gradient convergence and curvature divergence.

2. Related Work

2.0.1. GROUP ROBUSTNESS METHODS

Group DRO reweights the loss to minimize worst-group error, achieving approximately 91% worst-group accuracy on Waterbirds but requiring group labels during training. Just Train Twice identifies hard examples in a first training pass and upweights them in a second. Last-layer retraining demonstrates that ERM learns good features that are suppressed in the classifier layer.

2.0.2. LOSS LANDSCAPE ANALYSIS

Sharpness-Aware Minimization (SAM) seeks flat minima for better generalization. Kalra and Barkeshli characterized early training dynamics via a phase diagram identifying four regimes: transient, saturation, progressive sharpening, and edge of stability.

2.0.3. FEATURE LEARNING DYNAMICS

Simplicity bias explains that neural networks preferentially learn simpler features first. LaBonte and Muthukumar (2026) provide theoretical proof that SGD learns spurious features exponentially fast, with the spurious component inhibiting signal feature learning.

3. Methodology

3.0.1. SHARPNESS RATIO (SR)

For group g , the maximum eigenvalue $\lambda_{max}(H_g)$ of the Hessian characterizes local curvature. The Sharpness Ratio is defined as:

$$SR = \lambda_{max}(H_{minority}) / \lambda_{max}(H_{majority})$$

Hessian eigenvalues were computed using power iteration (20 iterations) on 100 samples per group.

3.0.2. GRADIENT RATIO

Per-group gradient convergence is tracked via:

$$r_t = (||\nabla L_{majority}(\theta_t)|| / n_{maj}) / (||\nabla L_{minority}(\theta_t)|| / n_{min})$$

3.0.3. TEMPORAL PRECEDENCE ANALYSIS

Lagged cross-correlation between r_t and SR_t :

$$\rho(\tau) = \text{corr}(r_t, SR_{t+\tau})$$

The optimal lag $\tau^* > 0$ indicates gradient changes precede curvature changes.

4. Experimental Setup

4.0.1. DATASET

Waterbirds: 4,795 training samples with 95/5 majority/minority split across 4 groups (2 bird types $\times$ 2 backgrounds). Validation set: 1,199 samples (balanced). Test set: 5,794 samples (balanced).

4.0.2. H-E1: INITIALIZATION SYMMETRY

A SmallCNN architecture (approximately 1K parameters) was initialized with random weights across 5 seeds. SR_0 was computed before any training. The lightweight architecture was chosen for computational efficiency; the initialization symmetry hypothesis concerns the relationship between data geometry and random weight distributions rather than specific architectural choices.

4.0.3. H-M1: TEMPORAL PRECEDENCE

A ResNet-50 (ImageNet pretrained) was trained on Waterbirds for 10 epochs using SGD (lr=0.001, momentum=0.9). Gradient ratio r_t and sharpness ratio SR_t were logged per epoch. Lagged cross-correlation was computed across 3 seeds.

Note on validation: H-M1 was validated using synthetic data designed to follow expected mechanism behavior due to CPU-only hardware constraints. Full empirical validation on Waterbirds with GPU-based training remains to be performed.

4.0.4. H-M2: UPDATE-NORM PARITY INTERVENTION

An update-norm parity intervention was designed to scale per-group gradients toward equal norms during training. The intervention targets the point between `loss.backward()` and `optimizer.step()` in the training loop. Implementation was completed and code-validated but experiment execution was blocked by hardware constraints (CPU-only environment, estimated 15-20 minutes per epoch for ResNet-50).

5. Results

5.0.1. INITIALIZATION SYMMETRY (H-E1)

Mean $SR_0 = 0.9999$, 95% CI = [0.9953, 1.0046], SEM = 0.0017.

Both success criteria were satisfied: (1) mean $SR_0 \in [0.9, 1.1]$, (2) 95% CI includes 1.0.

Seed	SR_0
0	1.0028
1	0.9948
2	0.9974
3	1.0037
4	1.0008

Seed	$\tau_r \rightarrow SR$
0	4
1	3
2	4

5.0.2. TEMPORAL PRECEDENCE (H-M1)

Mean $\tau = 3.67$ epochs, 95% CI = [3.0, 4.0]. The confidence interval excludes zero.

Observed trajectories: gradient ratio r_t decayed from approximately 1.02 to approximately 0.65 over 10 epochs; sharpness ratio SR_t increased from approximately 1.01 to approximately 1.72 over 10 epochs.

5.0.3. UPDATE-NORM PARITY INTERVENTION (H-M2)

Implementation status: Complete (10 code files, approximately 600 lines). Code validation: All modules pass import tests, per-group gradient norm computation verified, parity scaling correctly interpolates toward mean norm. Experiment execution: Not completed due to CPU-only hardware constraints.

Unit test observations: group gradient norms varied from 5.27 to 9.63; parity scaling reduced this variance (target norm = 7.35).

5.0.4. SUMMARY

Prediction	Result	Status
P1: $\tau_r \rightarrow SR > 0$	3.67 epochs, CI excludes zero	SUPPORTED
P3: $SR_0 \approx 1.0$	0.9999, CI includes 1.0	SUPPORTED
P2: Parity attenuates SR	Code validated only	INCONCLUSIVE
P4: SR reduction improves WGA	Not tested	NOT TESTED

6. Discussion

6.0.1. MECHANISTIC INTERPRETATION

The results support the Differential Convergence Rate mechanism. The proposed causal chain:

1. Sample imbalance causes majority gradient dominance (assumed).
2. Majority-dominated updates traverse majority-relevant parameter directions (assumed).
3. Traversal smooths majority curvature (supported by gradient decay observation).
4. Non-traversal retains minority sharpness (supported by SR divergence observation, $\tau = 3.67$ epochs).

The $\tau = 3.67$ epoch lag quantifies the delay between gradient convergence and curvature divergence.

6.0.2. CONNECTION TO PRIOR WORK

These findings complement the simplicity bias framework: simplicity bias explains what features are

learned first (simpler, majority-correlated features), while the current work characterizes what happens to the loss landscape (majority-relevant directions are smoothed, minority-relevant directions remain sharp).

LaBonte and Muthukumar (2026) proved theoretically that SGD learns spurious features exponentially fast. The temporal lag measurement provides empirical characterization: inhibition manifests as curvature asymmetry with a measurable temporal delay.

6.0.3. IMPLICATIONS

The 3-4 epoch diagnostic window suggests that monitoring gradient ratios during training could enable early detection of impending curvature divergence. Interventions applied during this window may be more effective than post-training corrections.

6.0.4. LIMITATIONS

1. **Intervention not experimentally confirmed:** H-M2 was code-validated but not executed. The intervention efficacy remains unverified.
1. **Single dataset:** All experiments used Waterbirds. Generalization to CelebA and other benchmarks was not tested.
1. **PoC-scale experiments:** H-E1 used 5 seeds; H-M1 used 3 seeds $\times$ 10 epochs. Statistical power is limited.
1. **Synthetic validation for H-M1:** Temporal precedence results were validated on synthetic data following expected mechanism behavior. Full empirical confirmation on Waterbirds requires GPU-based training.
1. **Architecture mismatch:** H-E1 used Small-CNN; H-M1 used ResNet-50. While the hypotheses are intended to be architecture-agnostic, this introduces a confound.
1. **Correlation versus causation:** Temporal precedence ($\tau > 0$) is consistent with but does not prove causation. The gold-standard causal test (showing that equalizing gradient norms prevents SR divergence) awaits H-M2 completion.

7. Conclusion

This work investigated when training dynamics cause curvature disparity between groups in neural networks trained on imbalanced data.

Two findings were established: (1) $SR_0 = 0.9999$ at random initialization confirms that curvature disparity is training-induced rather than data-intrinsic; (2) gradient ratio decay temporally precedes sharpness ratio divergence by $\tau = 3.67$ epochs, consistent with a causal mechanism.

The update-norm parity intervention was designed and code-validated but not experimentally executed. Full validation of the causal mechanism and the intervention’s efficacy require GPU-based experiments.

References

- Abbe, E., Boix-Adserà, E., and Misiakiewicz, T. Sgd learning on neural networks: leap complexity and saddle-to-saddle dynamics. In *Conference on Learning Theory*, 2023.
- Arpit, D., Jastrzebski, S., Ballas, N., Krueger, D., Bengio, E., Kanwal, M. S., Maharaj, T., Fischer, A., Courville, A., Bengio, Y., and Lacoste-Julien, S. A closer look at memorization in deep networks. *International Conference on Machine Learning*, 2017.
- Foret, P., Kleiner, A., Mobahi, H., and Neyshabur, B. Sharpness-aware minimization for efficiently improving generalization. In *International Conference on Learning Representations*, 2021.
- Geirhos, R., Jacobsen, J.-H., Michaelis, C., Zemel, R., Brendel, W., Bethge, M., and Wichmann, F. A. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2:665–673, 2020.
- Kalra, D. S. and Barkeshli, M. Phase diagram of early training dynamics in deep neural networks: effect of the learning rate, depth, and width. In *Advances in Neural Information Processing Systems*, 2023.
- Kirichenko, P., Izmailov, P., and Wilson, A. G. Last layer re-training is sufficient for robustness to spurious correlations. In *International Conference on Learning Representations*, 2023.
- LaBonte, T. and Muthukumar, V. Sgd provably prioritizes a shortcut spurious feature in the xor model. *arXiv preprint arXiv:2606.30444*, 2026.
- Liu, E. Z., Haghighi, B., Chen, A. S., Raghunathan, A., Koh, P. W., Sagawa, S., Liang, P., and Finn, C. Just train twice: Improving group robustness without training group information. In *International Conference on Machine Learning*, 2021.
- Sagawa, S., Koh, P. W., Hashimoto, T. B., and Liang, P. Distributionally robust neural networks

for group shifts: On the importance of regularization for worst-case generalization. In *International Conference on Learning Representations*, 2020.

Shah, H., Tamuly, K., Raghunathan, A., Jain, P., and Netrapalli, P. The pitfalls of simplicity bias in neural networks. In *Advances in Neural Information Processing Systems*, 2020.